

# Garrett Fincke

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## Education

### University of Pittsburgh

Aug 2026 – Expected May 2028

*Master of Science in Computer Science; SCI Scholarship for Excellence*

*Pittsburgh, PA*

### The Pennsylvania State University

Aug 2021 – Dec 2024

*Bachelor of Science in Computer Science*

*University Park, PA*

## Experience

### Westinghouse Electric Company

May 2026 – Aug 2026

*Software Developer Intern, GIC Tools and Applications*

*Cranberry Township, PA*

- Built a Python 3.14/FastAPI service exposing Oracle Document Index data through 15 OpenAPI-documented operations as a source-grounded retrieval surface for LLM and agentic workflows
- Built deterministic JSON, X-Request-ID correlation, server filters, hierarchy traversal, and CI reference resolution for an internal Databricks agentic AI platform
- Implemented WEC AD and project Oracle auth, rate limiting, failed-login throttling, error scrubbing, and readiness probes; validated across 500+ baselines/releases, 1,700+ locations, and 60,000 CI memberships

### Scale AI

May 2024 – Jul 2025

*Software Engineer (contract)*

*Remote*

- Built Python evaluation harnesses for LLM-generated code, compiling and executing submissions in sandboxed Docker environments and scoring correctness against automated tests
- Automated linting, test orchestration, and metrics pipelines to track pass@k, syntax/style errors, and failure modes across code-generation tasks
- Created schema-versioned data pipelines using Pydantic, JSON Schema, and deterministic sampling, replacing ad hoc manual review with reproducible scripted validation

## Projects

### SwimMate v2 | Python, Django REST, PostgreSQL/pgvector, OpenAI, Swift, SwiftUI, iOS/watchOS

Full-stack swimming platform with native iOS/watchOS apps and AI-generated workouts

- Built hybrid RAG workout generation over separate 768-dim builder and 1,536-dim retrieval paths with JWT auth, constraint extraction/reranking, role composition, provider abstraction, and audit logs
- Added offline 48-case eval harness with schema-versioned corpora, deterministic validation, hybrid/dense retrieval comparisons, latency/token metrics, and regression reports

### Coral | TypeScript, React Ink, Ollama, Node.js, SQLite, TypeScript Language Server

Local-first CLI/TUI coding agent powered by Ollama with semantic retrieval and code intelligence (v0.12.0)

- Built multi-turn agent loop with parallel tool use, read-only research subagents, and structured result feedback across file, shell, search, and Git operations
- Hardened local-model reliability with tool-call recovery, schema validation/coercion, doom-loop detection, SQLite semantic code search, and bundled TypeScript definitions, references, hover, and diagnostics
- Added session persistence, conversation compaction, and per-model context sizing for long-running tasks

### MDX Preview for VS Code | TypeScript, React 19, MDX 3, VS Code Extension API

Published VS Code/Open VSX extension (10,000+ installs, v1.6.1) with companion mdx-forge runtime tooling

- Architected a 6-package monorepo with React 19 webview and reusable mdx-forge compiler/runtime packages
- Built trust-gated Safe/Trusted rendering with DOMPurify, strict CSP, full React 19 components, 5 framework shims, 20 commands, and 32 settings

## Technical Skills

**Languages:** Python, TypeScript/JavaScript, Swift, Java, SQL, C/C++

**Backend & Data:** Django, FastAPI, Node.js, Postgres/pgvector, Oracle, Redis, Celery, Pydantic, REST

**AI/LLM:** RAG, embeddings/vector search, agentic AI, MCP, Ollama, vLLM, OpenAI/Anthropic APIs

**Frontend & Mobile:** React, Next.js, SwiftUI, UIKit, Tailwind CSS

**Cloud & DevOps:** Docker, Docker Compose, AWS EC2/SSM/CloudWatch, GitHub Actions, CI/CD, Vercel